Gyu Seon Kim

Ph.D., Aerospace AI Research Engineer

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Current Position

- Aerospace AI Research Engineer (Military Service Exception) at the R&D Center, RealtimeVisual Co., Ltd., Seoul, Republic of Korea

Research Interests

- **Autonomous Aerospace Systems:** Control and Communication for Autonomous Aerospace Systems (e.g., Aircraft, Drone, UAV, UAM, Fixed Wing, Rotary Wing, Rotorcraft, Launch Vehicles, and Spacecraft)
- **AI for Defense and Military Applications:** Intelligent Control and Autonomous Decision-Making for Defense and Aerospace Systems (e.g., Fighter Aircraft, Military Drone, Anti Drone System, Missile, Decoy, and Torpedo)
- **MARL-based Autonomous Control:** Multi-Agent Reinforcement Learning for Cooperative and Networked Aerial Vehicle Systems
- **RL-based Satellite Communication Systems:** Reinforcement Learning-based Communication and Networking for Satellite Systems (e.g., LEO Satellite Routing, Cube Satellite Design, Satellite Constellations, and Space-Air-Ground Integrated Networks)
- **Quantum AI:** Quantum Reinforcement Learning for Aerospace Control and Communication Systems

Education

Ph.D. **Korea University - College of Engineering**, Electrical and Computer Engineering (ECE) — Seoul, Republic of Korea

B.S. **Inha University - College of Engineering**, Aerospace Engineering — Incheon, Republic of Korea

Academic Background

- **Korea University - Research Institute of Information and Communication Technology**, Seoul, Republic of Korea - *Postdoctoral Researcher*
- **University of Houston - College of Engineering (ECE)**, Houston, TX, USA - *Visiting Scholar, Wireless Networking, Signal Processing and Security Laboratory (Prof. Zhu Han)*
- **Korea University - College of Engineering (ECE)**, Seoul, Republic of Korea - *Research Assistant, Artificial Intelligence and Mobility Laboratory (Advisor: Prof. Joongheon Kim)*

Industry Background

- **RealtimeVisual Co., Ltd. - R&D Center**, Seoul, Republic of Korea - *Aerospace AI Research Engineer (Military Service Exception)*
- Developing reinforcement learning-based flight-control algorithms for fighter aircraft and other aerial vehicles, with a focus on sim-to-real transfer of learned policies to physical flight platforms.

Awards and Honors

- **IEIE Best Paper Award** (November 2025) - *Lightweight Control Modeling via Quantum Reinforcement Learning*
- **IPESK Next Generation Engineering Researcher Award** (October 2025) - Funded by an Institute for Promotion of Engineering and Science of Korea
- **IEEE Vehicular Technology Society Student Scholarship Award** (October 2025) - Funded by an IEEE VTS Endowment Fund and partially funded by a VTS Award Fund, and administered by the IEEE Foundation
- **KICS Best Paper Award** (November 2024) - *Reinforcement Learning-Based Counter Measure Tactics to Avoid Torpedo Threats*
- **IEEE Seoul Section Student Paper Contest, Best Paper Award** (December 2023) - *Aircraft Taxi Routing using Reinforcement Learning at Hartsfield Jackson Atlanta International Airport*
- **KICS Best Paper Award** (February 2023) - *Self-Learning-Based Hybrid MAC for Military UAV Networks*

Professional Profiles

- Academic email: kingdom0545@korea.ac.kr
- Laboratory: Artificial Intelligence and Mobility Laboratory
- Google Scholar
- LinkedIn
- DBLP
- DBpia
- GitHub
- IEEE Xplore author profile

Professional References

- **Prof. Joongheon Kim**, Ph.D. Research and Dissertation Advisor - Professor, Korea University, Seoul, Republic of Korea; joongheon@korea.ac.kr Website
- **Prof. Zhu Han**, Ph.D. Visiting Research Supervisor - Professor, University of Houston, Houston, TX, USA; zhan2@uh.edu Laboratory website